Nikolas Varga

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EDUCATION

Northeastern University – Boston, MA December 2025

Bachelor of Science in Electrical Engineering and Music

Honors: 3.9/4.0 GPA, Dean's List, Professor Arvin Grabel Memorial Scholarship

Relevant Courses: Embedded Audio Programming*, Introduction to ML*, Electronic Design, Fundamentals of Electromagnetics, Fundamentals of Linear Systems, Acoustics and Psychoacoustics, Digital Design, Fundamentals of Electronics, Embedded Design

Activities: Eta Kappa Nu, Tau Beta Pi, Generate Product Development, Northeastern Acoustical Society of America (President), Northeastern Music Department Student Advisory Board, Green Line Records

TECHNICAL SKILLS

Hardware: Analog Filter Design, Buck/Boost Converter Design, Digital Logic Design, Discrete-time PID, Function Generator, HIL Testing, Soldering, Sensor Integration, Oscilloscope, VNA

Software: Altium Designer**,** AutoCAD, Bash, C++, Embedded Systems, Git, KiCAD, LTSpice, MATLAB, NI LabVIEW, OpenCV, Platform.io, Python, RISC-V Assembly, SystemVerilog

EXPERIENCE

Harvard Medical School, Garner Lab – Boston, MA July 2024 – December 2024

Computational Research Assistant

- Phase-matched 2 LEDs and 5 computer backlight signals to 2-photon laser sync signal to reduce 2-photon microscopy imaging noise by 100% using LabVIEW
- Designed 2-photon laser tracking system to maintain imaging location across experimental trials in LabVIEW
- Integrated LabVIEW auditory stimulus player with FSM, volume control, and buffer control with existing software, enabling place-preference tests in virtual 3D environments
- Applied cross-correlation processing techniques to analyze behavioral and auditory signals in MATLAB
- Refactored 3D rendering Python codebase to object-oriented architecture, implementing advanced features including 3D rotation, shearing, collision detection, and vector-based shape processing
- Studied auditory cognition and perception with interdisciplinary team of neuroscientists and engineers

Nano-C, Inc. – Westwood, MA July 2023 – December 2023

Device Lab Research and Development Co-op

- Fabricated and tested over 100 solution-processed organic photovoltaic devices and thin-film transistors in sustainable materials science setting to improve efficiency for future devices
- Formulated new ink solutions to optimize device performance for indoor and outdoor lighting applications
- Streamlined Device Lab Python analysis tools using boxplot, regression, and dataset manipulation features, leading to more significant statistical analysis and decreasing analysis times by 50%

PROJECTS

Giano – *Capstone Design* Fall 2025

- Built computer vision pipeline in Python/OpenCV using ArUco markers to correct perspective for mask generation
- Implemented FM-based multi-voice real-time harmony in embedded C++ to enable dynamic, realistic piano sound
- Designed 4-layer mixed-signal PCB for the glove, integrating Teensy 4.0, DRV2605L haptic drivers, and buffered tactile sensors with RC anti-alias filters to meet 75x50mm size, power, and mechanical constraints

Candlemaker – *Generate* Spring 2025

- Designed ESP32 controlled heating system using Kapton heaters and power MOSFET to melt candle wax
- Detected system temperature using thermistor sensor, controlling heating signal duty cycle using discrete time PID
- Tested and tuned PID in C++, implementing mechanical engineers' constraints and reducing overshoots by 33%

Tubender – *Generate* Fall 2024

- Designed 5V and 12V buck converter filter and feedback networks to power subsystems efficiently
- Layed out 4-layer PCB with ESP32, flash/boot circuitry, TFT connectors, and differential signaling in KiCAD
- Wrote embedded C++ to track ticks of rotary encoder, control multiplexer, and address I2C GPIO expander
- Managed version control and project collaboration on GitHub, enabling deep team integration with branching